ERC Consolidator Grant 2027

Research proposal [Part B1]

A New Web: Bridging NFPs for Heterogeneous Societal Requirements

ISLANDS

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Proposal duration: 60 months

Abstract

The World Wide Web scaled to billions of users through architectural decisions that gave up stable references, versioning, and bidirectional links. The downsides of this trade-off show today, when information conveyance has societal consequences: a fabricated audio clip of a party leader circulated, unverifiable, during Slovakia's 2023 pre-election silence; over half the URLs in UK parliamentary answers of 1997–2006 no longer resolved by 2007; courts in Europe and beyond have received over 1,200 filings with AI-fabricated citations; and copyright courts from Munich to New York must reconstruct attribution trails the Web never recorded. The Web's architecture hence deferred, rather than avoided, the cost of coordination, and society now pays it.

ISLANDS extends today's Web with an architecture organized around *non-functional properties* (NFPs). *Islands* are bounded networked information spaces whose members choose the NFPs their context demands and provide them jointly. Within an island, NFPs such as *persistence*, *confidentiality-preserving references*, and *attribution* are restored through *immutable versions*, *secure-multi-party-computation-based resolution*, and *traceable transclusion*. Across islands, we establish interoperability through the open Web's existing low-coordination high-scalability mechanisms, thereby also guaranteeing backwards-compatibility with today's World Wide Web. The scientific core of ISLANDS is a *theory of chosen guarantees* that predicts which NFP bundles an island of a given size, stake, and threat model can hold at what coordination cost. We develop this theory, embody it in a reference architecture, and validate it in legal citation, cross-organizational supply-chain, and online social media use cases, where our team has most experience. Our theory and open reference architecture will give society an extended Web whose guarantees are chosen by individual user communities that require them, and on which courts, parliaments, and markets can rely.

Cross-panel / interdisciplinary explanation

The proposal is computer-systems research and is submitted to **PE6 (Computer Science and Informatics)**. It spans four PE6 descriptors: PE6_10 (Web and information systems) as its core, PE6_2 (distributed systems), PE6_5 (cryptology, security, privacy; the confidential references and border checks of O3 and O4), and PE6_7 (artificial intelligence, intelligent systems, multi-agent systems; the agent layer of O4). We expect the scientific contribution to lie in the composition, which none of these sub-fields produces by itself.

Two validation domains and one objective also engage legal and governance scholarship (**SH2**, descriptor SH2_4, legal studies): we design the legal-citation island with legal practice and the social-media island against the data-access and transparency duties of the Digital Services Act, and the stakeholder analysis (O1, O5) builds on legal scholarship on architecture as regulation, on RFC 9518's centralization analysis, and on the PI's published work on DSA auditing and transparency regulation. SH2 expertise would sharpen the assessment of O1 and O5; the methods and deliverables are PE6, and SH2 is flagged as secondary.

Section a: Part I of the Scientific Proposal

b.1 The problem and the idea

The World Wide Web's original design favored simplicity, decentralization, and low coordination overhead, which enabled explosive adoption but sacrificed stable references, built-in versioning, and native bidirectional linking. This was a deliberate choice: Berners-Lee's 1989 proposal required that the system "allow existing systems to be linked together without requiring any central control or coordination" [1], and Fielding and Taylor later clarified the trade-offs that the resulting architecture yields when defining the Representational State Transfer (REST) architectural style: scalable component interactions, uniform interfaces, independently deployable components, and intermediaries that can be inserted anywhere [2]. These priorities suited early document-sharing systems and enabled the Web's scaling up to billions of users and devices within three decades. We argue that the Web's trade-offs are however too rigid for today's use cases, where publishers require traceable dereferencing, supply chain management requires aggregation over confidential data, courts and parliaments require stable, specific, citations, users require sovereignty over their data, and society requires trusted content and accountable authorship.

Over the past twenty years, we have kept the Web's architecture fundamentally static and such properties lacking; however, platforms have responded: Accounts, persistent storage, search, reputation, payment, and a consistent shared state do not exist between independent Web sites; platforms provide them inside proprietary stores into which they draw the Web's content, users, and interactions [3, 4, 5], and they extend into the rest of the Web through APIs and plug-ins on their own terms [6]. Users then rebuild continuity themselves through reposts, screenshots, and off-platform link hubs [7], yielding unsatisfactory mitigation. Berners-Lee warned in 2010 that social networks were walling off the information their users posted [8], and RFC 9518 describes the general mechanism: functions built on open protocols still centralize through network effects and switching costs [9]. The consequences reach the public sphere: Engagement-optimized ranking amplifies some political content over other content [10], and the EU's Digital Services Act now obliges the largest platforms to assess such systemic risks and to open their data to vetted researchers [11], an obligation that remains hard to operationalize because researchers must specify data whose existence they cannot observe [12]. The Commission's first fine under the Act, €120 million against X in December 2025, concerned a "verified" badge that anyone could buy without an identity check and terms that barred researchers from public data [13]. The Web itself authenticates domains through TLS certificates, but it offers no native way to state verifiably who stands behind a document or an account, or what qualifies them [14]. The W3C now lists among its Ethical Web Principles that the Web "makes it possible to verify information", since "the public needs verifiable source and context information to recognize trustworthy web publishers and content" [15]. Even the data that platforms do release differs from what their users see [16, 17]. Press publishers face the same opacity: in October 2026, the Paris Court of Appeal upheld an order that Meta negotiate neighbouring-rights payments with French media in good faith, finding that Meta had supplied only fragmentary, opaque, and unverifiable data on its use of their content [18]. Legal scholarship treats such architectural choices as regulation in their own right [19, 20, 21]. And, currently, generative AI repeats the pattern: answer engines and agentic browsers return answers in place of links and so control access to the Web's content [22], although the W3C Technical Architecture Group (TAG) holds that a user agent acting on its user's behalf owes "meaningful visibility into the sources it relied on" [23]. Highlighting real-world consequences, Reddit announced in October 2026 a move to a primarily logged-in experience, to protect its data from scraping [24] after AI answers cut its search referrals by more than a million per week. This shows that, today, the properties that high-stakes communities require are available, but inside closed platforms and under these platforms' control – hardly an Open Web.

Society pays for these omissions: In 2007 the UK government found that more than half of the roughly 4,000 URLs cited in ministerial answers to Parliamentary Questions since 1997 no longer resolved, and built a redirect service into its national Web archive in response [25]; one in five scholarly articles cites Web content that has since rotted [26]. Two days before Slovakia's 2023 election, a fabricated recording of a party leader discussing vote-rigging circulated with no provenance that voters could check [27]. From February 2027, batteries sold in the EU (and, soon after, products from clothes to detergents) must carry a digital passport with carbon-footprint and due-diligence data assembled from suppliers along the chain [28]. Law shows the cost most sharply, because there a reference's stability is part of the claim it supports. The EU has given judgments and legislation persistent identifiers through ELI [29, 30], but these name documents, not cited passages or versions; half the URLs cited in U.S. Supreme Court opinions no longer resolve, and Perma.cc archives cited pages one citation at a time [31]. By early 2026, more than 1,200 court decisions worldwide concerned AI-fabricated citations [32], and the High Court of England and Wales warned that citing fabricated authorities may lead to referral to the regulator or to contempt proceedings [33]. Content provenance is being moved into camera hardware [34] and into EU law [35, 11], and the Munich Regional Court held in November 2025 that a language model's memorization of song lyrics infringes copyright [36], a similar question to what *The New*

York Times v. OpenAI is litigating in the U.S. [37, 38]. What these cases have in common is that, in each case, coordination cost that the Web avoided was deferred, not eliminated; and society now pays it one incident at a time.

The idea. We propose to extend the Web's architecture to build selected properties into the Web itself, in a way that permits communities that require them to select their own property configuration, i.e., a set of desired *non-functional properties* (NFPs). ISLANDS introduces the *island* to capture such configured spaces: *Islands* are bounded networked information spaces, such as a legal-citation community, a supply-chain disclosure network, or a community of public-interest publishers on social media (Islands L, S, and M in Figure 1), whose members share stakes high enough to justify the cost of coordination through an ex-ante selected architectural trade-off rather than ex-post patching or litigation. Each island selects the NFPs its stakes demand and provides them jointly within its scope, while islands interoperate with one another and with the open Web through the Web's existing low-coordination mechanisms that permit planetary scaling. Within an island, references can for instance be defined as span-level, version-pinned, bidirectional, and verifiable, similar to how Xanadu specified them [39, 40]; across the boundary, they degrade to ordinary Web links, so that every island remains backwards-compatible with today's Web. Islands can also move away from the Web in the opposite direction: an island of fixed, high-speed installations may bind its interfaces once, at design time, in a binary encoding, much as a linker resolves a program's symbols when it builds the program [37, 38]. Entering an island is an explicit admission step: a client, human or agent, is shown the island's NFP set and accepts the obligations of membership, and this is where the island applies its attribution regime. Through ISLANDS, we further explore NFPs that have not yet been proposed on an architectural level. These include *confidentiality-preserving references* that resolve to a value that members can verify was correctly derived from data they may not read; this generalizes *secure multi-party computation* (SMPC), which has run in production since 2008 [41, 42]. They also include *purpose binding* which attaches the terms of a disclosure (recipient, purpose, validity window) to the reference, so that a reference is re-resolved when its terms expire [43]. *Attribution* binds a reference to an accountable author or, through anonymous credentials and the pseudonyms the European Digital Identity Wallet must support, to an eligible but unidentified one [44, 45, 46]. *Verifiable credentials* let a reference carry an issuer's attested claim about its source, e.g., that a filing was signed by a member of the bar or that an emissions figure was checked by an accredited verifier [14, 47]. *Capability discovery* shows agents, after each interaction, only the actions currently valid, in the hypermedia style of REST [48] as extended to multi-agent systems [49, 50, 51]; current LLM agents instead receive statically linked tool catalogues [52, 53]. The unifying scientific contribution is a *theory of chosen guarantees*: a formal account of which NFP bundles an island can provide, at which coordination cost, under which threat model, and for which stakeholders. Neither hypertext research nor Web engineering offers this today (b.2).

Figure 1 shows five example islands with different NFP selections, connected to each other through the open Web: legal citation and copyright (Island L), supply-chain disclosure (Island S), public-interest social media (Island M), scientific claims published as nanopublications (Island N), and fixed industrial installations (Island F). The theory predicts which selection an island of a given size, stake, and threat model can sustain.

- *Island L* lets a court or author cite a passage that resolves to exactly the version relied on, makes a fabricated citation fail visibly, and leaves a trail of reuse that a copyright dispute can follow; it does not establish that a cited source is correct, and it records no reuse that happens outside the island, such as the training of a language model on public copies.
- *Island S* lets suppliers report emission figures that buyers and accredited verifiers can check without the underlying production data leaving the supplier; it does not make a measured value accurate beyond what attested inputs cover, and it cannot compel a supplier outside the island to disclose.
- *Island M* lets readers see who stands behind an account and how a post was edited or deleted, and lets researchers obtain exposure statistics without raw-data transfer; it does not stop fabricated content from circulating on the open Web, and it makes provenance checkable, not content true.
- *Island N* keeps every scientific claim citable and attributed, with retractions on record; it does not verify that a claim is valid, nor tie it to the passage of the paper on which it rests, which is why WP1 asks whether this island also needs span addressing.
- *Island F* lets fixed installations interact over binary interfaces at low latency while staying described and reachable on the Web; it gives up evolvability, since any change on either side breaks a binding visibly and requires a new design-time agreement, and it provides none of the content guarantees of the other islands.

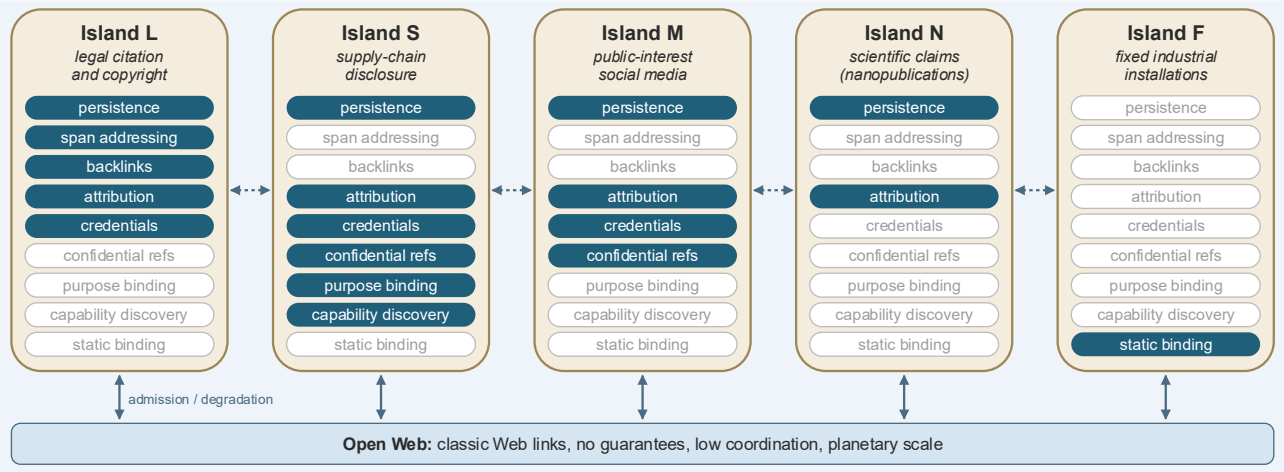


Figure 1. Five example islands with different NFP selections (filled: selected). These are the island drafts that the project solidifies and uses as requirements documents for its architecture: Island F, the downgrade island, binds its interfaces at design time (Static Links). Dashed arrows are references between islands; solid arrows are the border crossing, i.e., admission inbound and degradation to classic Web links outbound. The open Web gives no guarantees and connects all islands at low coordination cost.

b.2 State of knowledge and research gap

Hypertext research traditionally assumed rich, span-anchored links (e.g., [39, 54, 55]), and its agenda included versioning and computed structure (e.g., [56]): a link would persist; it would point at a specific range of bytes rather than a full page; and modifications of the link would lead to the creation of a new version rather than the replacement of the link. Open-hypermedia systems in addition implemented *bidirectional* links and *referential integrity* at workgroup scale [57, 58]; link rot had thus been addressed even before the Web made it endemic. The Web dropped these guarantees to scale [1, 48], and this was a sensible and pragmatic choice at the time.

Since the mid-1990s, when the Internet Archive began to archive the Web [59], remedial infrastructure (e.g., persistent identifier overlays, Web archives, Memento [60], Perma.cc [31], and robust hyperlinks [61]) has kept this architecture and compensated for its drawbacks, such as link rot and misattribution, e.g., through Web archives with permanent links such as the *Internet Archive's Wayback Machine* or, in the research space, *Zenodo* [62] (since link rot undermines reproducibility [26]). Such compensation did not touch the architectural principles of the Web, since changing them might fragment what the W3C's Ethical Web Principles call "one web" [15]. W3C specifications address passages and backlinks on the open Web: *Web Annotation* selectors [63] and *text fragments* [64] identify a passage of a page, and *Webmention* [65] and *Linked Data Notifications* [66], which the W3C *Social Web Working Group* maintains together with *ActivityPub* [67], deliver opt-in backlinks. None of them, however, pins a version or verifies what it points to, since this would induce architectural change with undesired side effects (e.g., it would undermine scalability of the Web). Dedicated services assemble specific NFP profiles as well, e.g., persistence via *Perma.cc* or *Zenodo*, provenance via *C2PA* [34], and identity and notarization via *DOIs* or *blockchains*. *Solid* [68] brings *data sovereignty*, since users keep their data in personal stores of their choice; *privacy and access control*, since each application reads only what the user grants it; and *portability*, since the data is decoupled from individual applications. These developments are evidence of the demand for chosen guarantees that extend beyond those of the basic Web architecture. However, each is an individual remedy that provides one fixed profile, and they do not touch the architectural foundation of the Web.

Today's re-decentralization efforts relocate *control* and leave *addressability* as it was. Solid's pods keep location-addressed, mutable resources [68], and data spaces relocate contracts and governance [69]. Federated social media behaves similarly: *ActivityPub* distributes servers but keeps posts location-addressed and mutable [67], and Bluesky's *AT Protocol* signs and content-addresses each user's repository [70] but leaves references between posts unverified and offers one fixed set of guarantees to all communities. Switching into the regulatory space, EU data law follows the same line as well: the *Data Act* and the portability rights of the *GDPR* and the *DMA* move *copies of data* between holders [71], and the Solid deployments that implement them (cf. our own [72, 73]) leave references location-addressed. However, raw-data transfer cannot sustain a market, hence we have recently seen proposals of trading *purpose-bound data derivations* whose terms limit their shelf life and use [43]. There is today no reference layer that could name such a derivation, pin it to a version, or carry its terms. *Usage control* formalized such terms [74, 75], and *sticky policies* bind them to the data [76]. Enforcement after release, however, requires a trusted monitor at the recipient or falls back on

detection after the fact [75]. RFC 9518 explains why open specifications alone do not prevent centralization, and we adopt the design-level mitigations it lists [9]. It is relevant to note that network architecture faced the same question one layer below. Clark et al. argued that an Internet embedded in society must leave room for the *tussle* between stakeholders with conflicting interests, and proposed design for change and controlled transparency as its tenets [77, 78]. Plutarch argued for heterogeneous network contexts joined by translating functions at their borders [79]. These positions state principles on the network layer that mirror the issues we highlight on the application layer of the World Wide Web, but they concern packets and network addresses rather than references and information resources.

As a community, we have thus built imperfect remedies instead of an architected approach. The reason is that an architected approach is hard: it requires consolidating fields that have so far advanced separately, namely hypertext, distributed systems, cryptography, multi-agent systems, and law.

We argue that, today, a widely applicable architectural shift is possible, based on methods that were not available in the early days of the Web: *Content-addressed* and *versioned storage* (Git; IPFS [80]; Software Heritage identifiers [81]) and *provenance attestation* [34, 82] now supply, at Web scale, the substrate that Xanadu [39, 40] lacked. They are, however, storage and signing primitives without span semantics, a link layer, or a governance model. *Nanopublications* [83, 84] form a deployed, federated network of immutable, signed semantic statements with self-verifying identifiers; they do not, however, offer span addressing over arbitrary content, *transclusion*, or confidentiality-preserving resolution; from the perspective of this proposal, nanopublications may become an *island* at statement granularity (Island N in Figure 1). With respect to who may see, trust, and act on the Web, too, the building blocks exist but are not part of the Web's architecture. With respect to confidentiality, communities need results computed over data that no single member may disclose. Secure multi-party computation (SMPC) has run in production since 2008 [41] and general-purpose protocols and implementations such as SPDZ [85] and MP-SPDZ [86] make computation practical. Our PEP-LCA protocol computes joint supply-chain assessments across up to 50,000 participants without a trusted third party [87], and a follow-on system anchors share disclosure in Solid's access-control layer [88]. Again, each of these deployments is, however, an individual application, and none exposes its result as a reference that others can cite and re-resolve. Confidentiality keeps data from the parties that use it; trust needs the converse, a verifiable statement about who stands behind the data. Verifiable credentials address this part of the problem: the W3C data model [14], selective-disclosure formats such as SD-JWT [89], and the European Digital Identity Wallet [46] let a holder prove an issuer's claim (e.g., a professional license) without revealing more than needed [47, 90]. They are, however, presented to one verifier within one session, and they are not bound to the references that the claim concerns. Credentials tell a client whom it deals with; autonomous agents also need to know what they may do. Hypermedia multi-agent systems research argues for run-time *affordance* discovery [49, 91]. Current LLM-agent practice, in contrast, uses statically linked tool lists [52, 53], which indirect prompt injection turns into an attack surface [92], and no existing design connects *capability discovery* to a verifiable reference layer.

The gap. We are not aware of any system that composes immutability, span-level verifiable addressing, transclusion, confidentiality-preserving resolution, and agent-facing affordance discovery *at the architectural level*. Closing this gap requires an understanding of each of these domains, spanning distributed computing, decentralization, and the societal requirements behind them. It must also be done in a backwards-compatible way that reuses the Web's architecture as deployed today and accepts its NFPs in bounded subsystems. Finally, for practical adoption, any such architecture needs a theory that predicts which NFP bundles fit a given deployment context and community.

b.3 Objectives

The goal of ISLANDS is to find out which guarantees a Web architecture can provide to the communities that need them, and at what cost, and to demonstrate these guarantees in practice. This involves:

- **O1 — Societally required islands.** We establish which *islands*, i.e., which bundles of NFPs, society requires today. We start from the five drafts of Figure 1: legal citation and copyright (Island L), supply-chain disclosure (Island S), public-interest social media that counters misinformation and manipulation (Island M), scientific claims published as nanopublications (Island N), and fixed industrial installations that trade the Web's uniform interface for binary interfaces (Island F). An *architectural decision record* states which islands the project does not pursue.
- **O2 — Theory of chosen guarantees.** We isolate the *link classes* that the required islands need and develop a formal model that relates them to coordination *and economic* costs, island size, threat models, and stakeholder incentives. The taxonomy has a persistence and integrity ladder (*Classic Jumplinks*, *Archived*, *Trusted*, *Verified*, and *Cryptographic Links*, in increasing strength) and five cross-cutting axes (attribution,

trust, usage, modality, *link time*). For each class, we prove what it minimally requires, and we derive decision procedures that an island designer can apply.

- **O3 — Border translation.** We determine how NFPs, and the concepts that support them (versions, identities, credentials, usage terms, encodings), are translated when information enters or leaves an island: which references are upgraded at admission, which are downgraded on exit, and how both happen without the silent loss of a guarantee.
- **O4 — Island technologies.** We build the technologies that the required islands need: a reference and *transclusion* layer over content-addressed storage; confidential references that turn SMPC-based resolution into an addressing primitive; and an agent layer with state-dependent *capability discovery* and design-time *Static Links*.
- **O5 — Validation and incentives.** We instantiate and evaluate Islands L, S, and M with their communities and Island F in industrial installations, and we measure whether the islands change each actor's incentive to centralize [9].

b.4 Research strategy and methods

The work packages follow the order of the argument: WP1 establishes the *islands* that society requires, WP2 and WP3 derive in parallel what these islands need inside and at their borders, WP4 builds the technologies, and WP5 validates them with their communities.

WP1 — Societally required islands (M1–M18, revised annually). Starting from the five drafts in Figure 1, we determine with the actors who would run and use them which NFP bundles society requires: in legal citation and copyright (Island L), supply-chain disclosure (Island S), public-interest social media (Island M), the scientific record, where the deployed nanopublication network [83, 84] is an island in all but name (Island N), and fixed industrial installations that deliberately downgrade the uniform interface to binary bindings (Island F). Interviews with each class of actor, the RFC 9518 checklist [9], and the W3C TAG's societal-impact self-review [93] test each draft. WP1 delivers an account of the required islands and an *architectural decision record* of the islands that ISLANDS does not pursue.

WP2 — Link classes and the theory of chosen guarantees (M6–M60). WP2 isolates the *link classes* that these islands need, from *Classic Jumplinks* to *Cryptographic Links*, with cross-cutting axes for attribution, trust, usage, modality, and *link time*, and formalizes them as a *guarantee calculus* in the information-flow control tradition. Each class carries two proof obligations, *realizability* and *necessity*, so that the calculus predicts which classes an island of given size, stake, and threat model can sustain, and at what cost.

WP3 — Border checks (M6–M48, in parallel with WP2). WP3 determines how NFPs, and the concepts that support them, are translated when information enters or leaves an island: which references are upgraded at admission, which are downgraded on exit, and which cannot cross. Its governing rule is that no reference loses a guarantee silently. WP3 delivers a translation table, an *admission protocol* based on *verifiable credentials* [14], and border gateways for each island.

WP4 — Island technologies (M12–M54). From the islands and their link classes, we build the technologies they require: a reference and *transclusion* layer over content-addressed storage (T4.1), confidential references that generalize our SMPC system [87, 88] into an addressing primitive (T4.2), and an agent layer with state-dependent *capability discovery* and design-time *Static Links* (T4.3).

WP5 — Validation islands and incentives (M18–M60). We instantiate Islands L, S, and M with their communities and demonstrate Island F in industrial installations, measure which guarantees each island holds and at what cost, and repeat the WP1 interviews to test whether the islands changed each actor's incentive to centralize.

Integration and openness. Annual integration milestones bind the prototypes into one architecture and revise the WP1 account and the calculus. All software and publications are open, and all specifications go to the W3C groups that maintain the protocols islands build on.

b.5 Team

ISLANDS consolidates the PI's research lines in Web architecture, ubiquitous computing, human-computer interaction, and law. In *Web architecture*, the Web of Things [94], its W3C standardization [95] (now deployed in industrial products), and hypermedia multi-agent systems [49, 50, 91] ground the *link classes* and border checks (WP2, WP3) and the *island* technologies (WP4). In *ubiquitous computing*, references that bind physical artifacts to digital records [96] and personalization systems that mediate what people perceive in their surroundings [97] extend *islands* from documents to devices, places, and the people in them. In *human-computer interaction*, the PI's work on *affordances* and *signifiers* [51] informs WP4's *capability discovery* and WP3's admission step at an island's border, where a client is shown the obligations it accepts. In *law*, regulation

expressed as technical architecture [21, 98, 99], platform auditing under the DSA [16, 17], and membership in the European VLOP-auditing oversight board ground Islands L and M, the attribution regime, and the stakeholder analysis (WP1, WP5). As General Chair of the Solid Symposium 2027 and of the ACM Symposium on Computer Science and Law 2028, the PI convenes the decentralized-data and the computer science and law communities on which WP1, WP4, and Island L draw. Each line has so far addressed one part of the problem. The *theory of chosen guarantees* (O2) needs all of them, because an island's guarantees are at once architectural, experienced by people, and legally consequential. The PI stepped down as Dean in February 2026 to pursue this programme (≥40% commitment).

Two team members carry the running systems on which the project builds. **Luka Bekavac** (through 2029, ≈M20) brings the empirical and regulatory instruments (e.g., platform transparency, DSA data-access auditing, integrity-by-design referencing [16, 17, 96]) for WP1's island requirements and stakeholder analysis and for WP5's legal and social-media islands. **Jan Grau** (through 2029, ≈M20) built the PEP-LCA SMPC stack and its Solid integration [87, 88], which anchor WP4's confidential references and Island S. We will recruit an applied-cryptography postdoc (confidential references in WP4 and border conditions in WP3, from M1), two PhD students (reference layer and border checks, WP3 and WP4, M1; agents and static bindings, WP4, ~M6), a socio-technical postdoc who succeeds Bekavac with overlap (WP1 and WP5, from ~M17), a further PhD student on the legal island (WP1 and WP5, from ~M21, co-supervised with the legal partner), and a research engineer for the shared prototype. External collaborators are Mitrokotsa and Cheng (cryptography), Ciortea (hypermedia MAS, USG), Tamò-Larrieux (law, Lausanne), the ODI and the Solid community (via CoCoDa), and the industrial WISER partners (Empa, Siemens).

b.6 Ambition, risk, gain

ISLANDS tests the hypotheses that a coordination cost the Web's designers judged prohibitive [1, 2] is sustainable at *island* scale, that cryptographic reference resolution can reach interactive latency, and that actors with divergent incentives will adopt shared guarantees. Each hypothesis can be refuted. WP1 hence asks which guarantees society requires, O2 treats them as *chosen*, and O5 measures incentives alongside latency. If our hypotheses hold, we will deliver a validated architecture for the parts of the information space where the Web's founding trade-off now fails at measurable cost, chiefly the parliamentary, legal, and scientific record, public discourse on social media, and regulated industrial data. Autonomous agents could then act on verifiable information in these spaces. If our hypotheses fail, the theory still delimits which guarantees are attainable, for whom, and at what cost.

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Section b: Curriculum Vitae and Track Record

Personal information

Prof. Dr. Simon Mayer; born 16 July 1987; Austrian; three children (2015, 2017, 2019). ORCID 0000-0001-6367-3454.

PhD: Dr.sc. ETH in Computer Science, ETH Zurich; **defense 12 September 2014** (the defense date counts); **eligible** for CoG 2027 under its 5–15-year window (12.3 years on 01.01.2027), no extensions needed. Further degrees: MAS in Secondary and Higher Education, ETH Zurich (2014); B.A. in Economics and Business Administration, University of Zurich (2014); M.Sc. ETH in Computer Science (2010).

Current and previous positions

- 2018–today: **Full Professor of Computer Science** (Interaction- and Communication-based Systems), University of St. Gallen (USG); one of the four founding professors of the School of Computer Science.
- 2024–02/2026: **Dean**, School of Computer Science, University of St. Gallen (stepped down to refocus on research, which frees the time for the ERC ≥40% commitment). 2022–2024 Academic Director; 2018–today Director and 2018–2024 Managing Director, Institute of Computer Science.
- 2020–today: Board Member, **Research Center for Information Law**, USG.
- 2023: Visiting Professor, University of Bologna. 2018–2020: Senior Scientist, Distributed Systems Group, ETH Zurich. 2017–2018: Area Manager "Cognitive Products", Pro2Future GmbH & Project Scientist, Graz University of Technology. 2014–2017: Research Scientist, then Senior Researcher and **Senior Key Expert, Siemens Corporate Technology**, Berkeley, USA. 2015–2017: Visiting Lecturer, UC Berkeley and ETH Zurich. 2013: Visiting Researcher, Ghent University. 2012: Visiting Researcher, MIT.

Ten most significant research outputs

(Chosen to evidence what ISLANDS needs: carrying Web architecture research into standards and products; law/technology co-design; and the two running systems the project builds on.)

1. **W3C Web of Things Thing Description standards**: the PI's PhD and Siemens-era research contributed to the basis of W3C WoT TDs, today deployed in industrial products (e.g., Siemens Desigo building management). Evidence for the exact pathway ISLANDS plans for its island specifications: research results carried into an open standard and from there into practice.
2. Ciortea, **Mayer**, Gandon, Boissier, Ricci, Zimmermann. *A Decade in Hindsight: The Missing Bridge Between Multi-Agent Systems and the World Wide Web.* AAMAS 2019. (Conceptual foundation of WP4; consolidated via two co-organized Dagstuhl seminars, 21072 & 23081, and the PI's founding membership of the W3C Web Agents CG.)
3. Vachtsevanou, Ciortea, **Mayer**, Lemée. *Signifiers as a First-class Abstraction in Hypermedia Multi-Agent Systems.* AAMAS 2023; and Vachtsevanou et al., *Enabling BDI Agents to Reason on a Dynamic Action Repertoire in Hypermedia Environments.* AAMAS 2024. (Run-time affordance exposure; WP4's direct technical basis.)
4. Guitton, Tamò-Larrieux, **Mayer**. ***AI and Law: How Automation is Changing the Law.*** Routledge, 2025. (Book consolidating seven years of the automatically-processable-regulation line; the documented legal-field record the motivation leans on.)
5. Tamò-Larrieux, Guitton, **Mayer**. *A Computational Science Perspective on the Legal System.* **Nature Computational Science**, 2025.
6. **Mayer**, Fuchs, Brügger, Lian, Ciortea. *Improving Customer Decisions in Web-based E-commerce through Guerrilla Modding.* **Nature Machine Intelligence** 3(12), 2021. (User-side re-engineering of Web experiences; an early form of the islands-vs-platform-defaults argument.)
7. Grau, Maftun, Garcia, **Mayer**. *Solid and Secure Multi-Party Computation for a Circular Economy.* Solid Symposium 2026; with Grau's MSc thesis (2025, supervised by the PI with Garcia, Mitrokotsa, Cheng). (Running system behind WP4/Island S.)
8. Bekavac, **Mayer**. *Platforms' Research API Data Access: What Users See vs. What Researchers can Retrieve.* ACM FAccT 2026 (preprint: arXiv:2601.12390); and Bekavac, Strecker-Bischoff, Garcia, **Mayer**, Tamò-Larrieux. *Scrutinizing Systemic Risks in Personalized Recommender Systems Through Sock-Puppet Auditing of VLOPs.* ACM Transactions on Recommender Systems 5(1), 2026. (Empirical basis of the stakeholder analysis, WP1 and WP5.)

9. Tamò-Larrieux, **Mayer**, Zihlmann. *Not Hardcoding but Softcoding Data Protection.* Technology and Regulation, 2021; with the *Artificial Intelligence and Law* line on automatically processable regulation (2022–2025). (Law expressed as architecture: the design stance ISLANDS generalizes.)
10. Two CHI 2025 **Honorable Mentions** (adaptive industrial robots; responsible ubiquitous personalization, DIS 2025) and U.S. Patent US11331803B2 (mixed-reality robot programming, with Siemens). (Breadth: HCI venues + transfer to IP.)

Funding record

PI or Co-PI on **4 SNSF, 2 Innosuisse, 3 EU, and 12+ industry/foundation projects** (3.9 MCHF allocated to the PI's group out of 13.6 MCHF total), including: SNSF **HyperAgents** (2019–2024, hypermedia MAS; WP4 ancestry); EU **IntellIoT** RIA (2020–2024); EU **AFarEdge** Chips JU RIA (2026–2029, task leader for the architecture; WP2/WP4 ancestry); Innosuisse Flagship **WISER** (2022–2026, climate intelligence/LCA; Island S ancestry); SNSF **CoCoDa** (2025–2029, VLOP regulation, with Lausanne, Maastricht, and the Open Data Institute; Island L and incentive-analysis ancestry); SNSF/Innosuisse Bridge Discovery **Access to Justice** (2026–2030, with Caritas Switzerland); TA-SWISS *Sprachmodelle und Kompetenzverschiebungen* (2025–2026).

Supervision and team-building

Built a research group and, as founding professor and Dean, a department (10 research groups, 120+ employees, ~300 students, within budget). Five postdocs mentored into faculty/senior positions (Gómez, now Assoc. Prof., TU Braunschweig; Garcia, TT Asst. Prof., IT:U Linz; Ciortea, Asst. Prof., USG; Guitton, Swiss Re; Bektaş, Senior Researcher, KSSG). PhD students: three graduated 2024/25 (Ramanathan, Hostettler, Jha), one completing (Vachtsevanou), six current (including Bekavac and Grau, both on this project's ancestor systems).

Service, standards, recognition

General Chair, **Solid Symposium 2027** and **ACM Symposium on Computer Science and Law (CS&Law) 2028**. Associate Editor, **PACM IMWUT** (since 2021); Editorial Board, IEEE Pervasive Computing (since 2019); Exec. Editor, Wiley Internet Technology Letters (2017–2023); SI Editor, ACM TOIT (2023). Steering Committee, International Conference on the Internet of Things (2017–); co-organizer, International Workshop on Hypermedia Multi-Agent Systems (2019–2020) and Web of Things workshop series (2012–2017); Dagstuhl seminars 21072 and 23081. Active in **W3C and IETF** standardization; founding member, W3C Web Agents Community Group. Member of the **European oversight board on Very Large Online Platform auditing** (to the PI's knowledge, its only computer scientist). Finalist, Best Teaching Award, USG (2024).

Track-record narrative

ISLANDS consolidates four research lines the PI has built separately: Web architecture, ubiquitous computing, human-computer interaction, and law. Ubiquitous computing (Associate Editor, PACM IMWUT; Steering Committee, International Conference on the Internet of Things) and HCI (two CHI 2025 Honorable Mentions; signifiers carried from HCI to multi-agent systems) connect the two lines described below, carrying Web architecture into physical environments and to the people who act in them. The Web-architecture line built systems that let machines use the Web: from Web of Things architecture (PhD, ETH; Siemens) through its W3C standardization and industrial deployment, to hypermedia multi-agent systems and run-time affordance discovery (AAMAS 2019–2024). This line supplies the architectural competence that the reference and agent layers (WP4) require, with a demonstrated record of moving designs of this kind into open standards and products, and the record is current: the PI presently leads the definition of the architecture in a €27.6M Chips JU action, where a Web-of-Things-based specification authored by the PI's group is the reference that over 1,500 person-months of hardware and platform implementation must satisfy. The legal line studied, with legal scholars, how information control works under regulation: automatically processable regulation, AI-and-law (Routledge 2025; Nature Computational Science 2025), DSA data-access auditing, and service on the European VLOP-auditing oversight board. This line supplies the domain competence Island L and the incentive analysis (WP5) require, and it backs the proposal's legally anchored motivation with the PI's own publications. The two running systems the project extends (the PEP-LCA/ Solid SMPC stack; the signifier-based agent environments) were both built in the PI's group by team members who carry them through the project's first 20 months.